WORKSHEET

Chapter 7: Partial fractions

1 Show that $\frac{2x+1}{x^2+x} = \frac{1}{x} + \frac{1}{x+1}$

2 Show that $\frac{4}{x^2-1} = \frac{2}{x-1} + \frac{-2}{x+1}$

3 Show that $\frac{x^2+7x-2}{x^3-x} = \frac{2}{x} + \frac{3}{x-1} + \frac{-4}{x+1}$

4 Show that $\frac{2x^3+3x-7}{x^3-3x^2+x-3} = \frac{2}{x-3} + \frac{3}{x^2+1}$

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5 Express the following in partial fraction form.

a $\dfrac{x^2}{x-1}$

b $\dfrac{x-8}{(x+2)(x-3)}$

c $\dfrac{x}{x^2-4x-5}$

d $\dfrac{11x-10}{x^2-2x}$

e $\dfrac{x}{(x+2)(x+3)}$

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f $\dfrac{1}{x^3 + x^2 + x}$

6 Express the following in partial fraction form.

a $\dfrac{x + 7}{(x + 2)(x - 3)}$

b $\dfrac{x^2 + 2}{x(x + 2)(x - 1)}$

c $\dfrac{x + 3}{x^3 - 4x}$

d $\dfrac{x^2}{x+1}$

e $\dfrac{2x-3}{(x+1)(x-4)}$

f $\dfrac{2x+1}{x^2+2x+1}$

7 Find:

a $\displaystyle\int \frac{2x-3}{(x+1)(x-4)}\,dx$

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b $\int \frac{x+7}{(x+2)(x-3)}dx$

c $\int \frac{x^2}{x-1}dx$

d $\int \frac{1}{x^2-4}dx$

e $\int \frac{x+1}{x^3+x^2-6x}dx$

f $\int \frac{x}{(x-2)^2}dx$

8 Show that $\frac{2x^3 + 3x - 7}{x^3 - 3x^2 + x - 3} = \frac{2}{x - 3} + \frac{3}{x^2 + 1}$ and hence find $\int \frac{2x^3 + 3x - 7}{x^3 - 3x^2 + x - 3} dx$.

9 Evaluate the following.

a $\int_0^2 \frac{x^2}{x^2 + 3x + 2} dx$

b $\int_1^3 \frac{x^2 - x}{(x + 3)(x + 1)} dx$

c $\int_1^2 \frac{x + 2}{x(x + 4)} dx$

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d $\int_{-1}^{1} \frac{x^2 + 3x - 2}{x^2 - 4} dx$

e $\int_{3}^{4} \frac{x + 7}{(x + 3)(x - 1)} dx$

f $\int_{-1}^{2} \frac{x^2 + 3x - 4}{(x - 4)(x + 2)} dx$

10 Show that $\int \frac{x^4 - x^3 - x - 1}{x^3 - x^2} dx = \frac{x^2}{2} + 2\ln|x| - 2\ln|x - 1| - \frac{1}{x} + c$

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Answers

1–4 Demonstrations

5 **a** $x+1+\dfrac{1}{x-1}$

b $\dfrac{2}{x+2}+\dfrac{-1}{x-3}$

c $\dfrac{5}{6(x-5)}+\dfrac{1}{6(x+1)}$

d $\dfrac{5}{x}+\dfrac{6}{x-2}$

e $\dfrac{-2}{x+2}+\dfrac{3}{x+3}$

f $\dfrac{1}{x}-\dfrac{x+1}{x^2+x+1}$

6 **a** $\dfrac{-2}{x+2}+\dfrac{3}{x+3}$

b $\dfrac{-1}{x+2}+\dfrac{1}{x-3}$

c $\dfrac{-3}{4x}+\dfrac{1}{8(x+2)}+\dfrac{5}{8(x-2)}$

d $x-1+\dfrac{1}{x+1}$

e $\dfrac{1}{x+1}+\dfrac{1}{x-4}$

f $\dfrac{2}{x+1}-\dfrac{1}{(x-1)^2}$

7 **a** $\ln\left(|x+1|\,|x-4|\right)+c$

b $\ln\left[\dfrac{(x-3)^2}{|x+2|}\right]+c$

c $\dfrac{x^2}{2}+x+\ln|x-1|+c$

d $\dfrac{1}{4}\ln\left|\dfrac{x-2}{x+2}\right|+c$

e $\dfrac{-1}{6}\ln|x|+\dfrac{3}{10}\ln|x-2|-\dfrac{2}{15}\ln|x+3|+c$

f $\ln|x-2|-\dfrac{2}{x-2}+c$

8 $2\ln|x-3|+3\arctan(x)+c$

9 **a** $\ln\left(\dfrac{3}{16}\right)+2$

b $\ln\left(\dfrac{128}{729}\right)+2$

c $\dfrac{1}{2}\ln\left(\dfrac{12}{5}\right)$

d $2-\ln(3)$

e $\ln\left(\dfrac{27}{14}\right)$

f $2\ln\left(\dfrac{8}{25}\right)+3$

10 Demonstration

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